QP CODE: 24900183

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOV 2024

First Semester

Discipline Specific Core Course - MG1DSCMOS100 - PC HARDWARE AND SMARTPHONE TROUBLESHOOTING

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Part A

Multiple Choice Questions Answer all questions. Each question carries 1 mark

1.	b) RAM	[U]	[1]
2.	b) Holds the CPU	[U]	[1]
3.	b) Process instructions	[K]	[2]
4.	a) Escalate to a more advanced support team.	[A]	[2]
5.	b) Shorting the green and black wire	[U]	[2]
6.	c) The boot mode is set incorrectly	[A]	[2]
7.	b) To provide a visual guide for decision-making	[U]	[2]
8.	c) GPU	[K]	[3]
9.	b) By regulating power consumption for various components.	[U]	[3]
10.	b) Accelerometer and gyroscope	[A]	[2]

Part B

Short Answer Questions Answer 4 questions. Each question carries 5 marks

11 Draw the block diagram of a PC and give its working (5 Marks) **[K] [1]**

Block Diagram (2 Mark):Components: Input devices, Output devices, CPU (ALU, CU, Registers), Memory (Primary and Secondary), Motherboard, PSU.

Explanation of Working (3 Marks):

- Input: Devices send data to CPU.
- Processing: CPU performs operations with data stored in RAM.

- Output: Processed data is sent to output devices.
- Storage: Data is saved to HDD/SSD.
- Power: PSU supplies stable power to components.

Reference Text: Upgrading and repairing of PC's- Scott Mueller's

12 Explain how the voltage regulator, BIOS ROM, CMOS battery, and RAM work together to ensure the smooth operation of a computer (5 Marks) **[U] [1]**

- **Voltage Regulator (1 Mark):** Stabilizes voltage levels for critical components.
- **BIOS ROM (1 Mark):** Contains firmware to perform POST (Power-On Self-Test) and load the OS.
- **CMOS Battery (1 Mark):** Powers CMOS to retain system time and BIOS settings.
- **RAM (2 Marks):** Temporarily stores data for the CPU during operation to enable multitasking.

Reference Text: Upgrading and repairing of PC's- Scott Mueller's

13 What is the purpose of the RF unit on a smartphone motherboard? (5 Marks) **[K] [2]**

RF Unit Purpose:

- **Enables wireless communication (2 Marks):** Facilitates cellular, Wi-Fi, Bluetooth, and GPS signals.
- **Signal Conversion (3 Marks):** Converts analog RF signals to digital and vice versa for processing.

Reference Text: Smartphone troubleshooting & repair

14 Name three common types of connectors found on the back panel of a motherboard (5 Marks)

[U] [2]

Common Types -any 3-5 marks, any 2-3 marks, any 1-1.5 marks

- USB (Universal Serial Bus) Ports: For peripheral connectivity.
- HDMI (High-Definition Multimedia Interface): For video and audio output.
- Ethernet Port: For network connectivity.

Reference Text: PC hardware maintanence & trouble shooting-N S Reddy

15 Compare the functions of the accelerometer and gyroscope in a smartphone (5 Marks) **[U] [2]**

- **Accelerometer (2.5 Marks):**Measures linear acceleration of the device.Detects orientation changes
- **Gyroscope (2.5 Marks):**Measures angular velocity.Provides precision in tracking rotation

Reference Text: Smartphone troubleshooting & repair.

16 Describe how GPS (Global Positioning System) functions in a smartphone (5 Marks) **[U] [2]**

- **Satellite Communication (3 Marks):** GPS uses signals from satellites to determine location.
- **Output (2 Mark):** Location data is used in navigation apps.

Reference Text: Smartphone troubleshooting & repair

Part C

Essay Questions Answer 2 questions. Each question carries 10 marks

17 Explain the working principle of a Switched Mode Power Supply (SMPS) in a PC (10 Marks) **[U] [1]**

Definition (2 Mark):SMPS converts AC to regulated DC for PC components with high efficiency.

Working Principle (7 Marks):

- Input Stage: Converts AC to DC using rectifiers and filters.
- Switching Stage: High-frequency switching by transistors (MOSFETs).
- Transformer Stage: Steps down voltage using a high-frequency transformer.
- Output Stage: Converts high-frequency AC back to DC and stabilizes using filters.

Advantages (1 Marks):High efficiency, lightweight, reduced heat dissipation.

Reference Text: Upgrading and repairing of PC's- Scott Mueller's

18 What is a chipset? Explain the working of each component of a chipset (10 Marks) **[U] [2]**

Definition (2 Marks):A chipset is a set of electronic components on a motherboard managing communication between CPU, memory, and peripherals.

Components of Chipset (6 Marks):

- Northbridge: Manages high-speed components like CPU, RAM, and GPU.
- Southbridge: Handles I/O devices, storage, USB, and audio.
- Clock Generator: Maintains system timing.
- Voltage Regulators: Provide power stability.

Working (2 Marks):Coordinates data flow between CPU, memory, and peripherals.

Reference Text: Upgrading and repairing of PC's- Scott Mueller's

19 Explain the role of the RF Unit in smartphones and discuss the types of antennas used in smartphones (10 Marks) **[A] [2]**

Role of RF Unit (5 Marks):

- Manages wireless communication (cellular, Wi-Fi, Bluetooth, GPS).
- Converts analog RF signals to digital for processing.
- Includes components like power amplifiers, filters, and modulators.

Types of Antennas (5 Marks):

- Cellular Antenna:Facilitate communication with cellular network for voice and data services
- WiFi Antenna: Enables wireless internet connection
- Bluetooth Antenna: allow the short range wireless communication with peripherals
- GPS antenna :Receive signal from GPS satellite for location service.

Reference Text: Smartphone troubleshooting & repair

20 Draw and explain the typical functional diagram of a smartphone (10 Marks) **[U] [3]**

Diagram (3 Marks):

Key Components:CPU,GPU,Memory (RAM, ROM, Flash Storage),RF Unit, Display and Touch Interface,Battery Management System,Sensors (Accelerometer, Gyroscope, Proximity)

Explanation (7 Marks):

- CPU and Memory: Perform data processing and storage.
- GPU: Handles the rendering of image,video & Graphical User Interface element
- RF Unit: Manages wireless communication.
- Battery Management: Ensures efficient power usage.
- Sensors: Enable features like orientation, motion detection, and proximity.

Reference Text: Smartphone troubleshooting & repair